

## The number $e \approx 0.1.5$

Define  $e$  to be the number  $\lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$ .

Suppose I have some amount  $A$  in an account and an interest rate of  $r$ .

After 1 year, the amount in the account is  $A + Ar = A(1+r)$ .

If instead the interest is given in installments throughout the year, I get more.

If interest is compounded twice in a year, then after  $\frac{1}{2}$  yr, there is

$$A + A\frac{r}{2} = A(1 + \frac{r}{2}) \quad \text{Then after 1 yr,}$$

$$\begin{aligned} \text{there is } & A(1 + \frac{r}{2}) + A(1 + \frac{r}{2})\frac{r}{2} \\ & = A(1 + \frac{r}{2})(1 + \frac{r}{2}) = A(1 + \frac{r}{2})^2 \end{aligned}$$

If interest is compounded  $n$  times throughout the year, then at the end of 1 yr, the account has  $A(1 + \frac{r}{n})^n$ .

Define  $x = \frac{n}{r}$  so that  $n = xr$ . Then

$$A(1 + \frac{r}{n})^n = A(1 + \frac{1}{x})^{xr} = A((1 + \frac{1}{x})^x)^r$$

Then as  $n \rightarrow \infty$ ,  $x$  also approaches  $\infty$

$$\text{and } \lim_{n \rightarrow \infty} A \left(1 + \frac{r}{n}\right)^n = \lim_{x \rightarrow \infty} A \left(1 + \frac{1}{x}\right)^x{}^r$$

$$= A \left( \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \right)^r = A e^r.$$

If we compound "continuously", then at the end of the year, the account has  $Ae^r$  dollars.